PYTHON TURTLE

Colourful Spiral Fun! 🚀

What is Python?

Python is a language computers understand to do fun things! We can tell computers exactly what to do by writing Python code.

Meet Turtle Graphics!

Turtle is like a **magical pen** that draws on the screen! It follows your instructions and leaves a colourful trail as it moves around!

Drawing with Turtle Graphics!

Move Forward & Turn

Turtle can move forward and turn left or right. Each move takes it to a new spot on the screen!

Draws Lines as It Moves

As Turtle walks across the screen, it leaves a line behind — just like drawing with a pen on paper!

Change the Pen Color!

We can tell Turtle which color to use. Try red, blue, green... and make your drawing super colorful!

What is a Spiral?

A spiral is a curve that goes round and round, getting bigger and bigger! It starts small in the middle and grows outward. We will draw one like this using Python Turtle!

STEP-BY-STEP CODING GUIDE

Step 1: Import Turtle

Type this first line in your code editor:

```
import turtle
```

This tells Python to bring in and use the Turtle drawing tool. Now we are ready to draw!

Step 2: Create the Turtle!

We make a turtle named `t` that will draw for us!

```
t = turtle.Turtle()
```

This creates your very own turtle ready to draw colourful spirals on screen!

Step 3: Make It Colourful!

We choose a list of colors for our spiral. Each time the turtle moves, it picks the next color from this list, making our spiral bright and beautiful!

```
colors = ['red', 'orange', 'yellow', 'green', 'blue', 'purple']
```

Step 4: Draw the Spiral Loop!

```
for x in range(100):
    t.pencolor(colors[x % 6])
```

```
    t.forward(x * 2)
    t.right(59)
```

How the Loop Works:

- **Loop repeats 100 times:** It builds the spiral piece by piece.
- **Pen changes color:** It cycles through our list using $x \% 6$.
- **Turtle moves forward:** It moves a little more each step ($x * 2$) so the spiral grows!
- **Turtle turns right:** It turns $59°$ to create the twisting spiral shape!

WATCH THE SPIRAL GROW!

What Did We Learn?

1. Importing Modules

We use `import turtle` to tell Python to bring in the Turtle drawing tool before we start coding.

2. Making a Turtle

We created a turtle named `t` using `turtle.Turtle()` — our magical pen that draws colourful spirals on screen!

3. Loops & Colors

We used a `for` loop to repeat drawing 100 times and a colors list to change the pen color each turn!

🎨 Try Your Own Spiral!

- Change the colors in the list!
- Try turning angles like *60°*, *90°*, or *45°*.
- Change how far the turtle moves each time.

Make it YOUR unique spiral!

📝 Let's Practice Together!

Can you draw your own colourful spiral today?

Try writing the code in an editor, or sketch your own beautiful spiral design on paper and share it with your classmates!

🛠️ Troubleshooting Tips & Help!

- **Check Your Spelling:** Python is very careful about spelling! Make sure every word in your code is exactly right.
- **Indent Your Loop:** The lines inside your loop must be indented (moved in with spaces or a tab). Python needs this to know what to repeat!
- **Ask a Friend or Teacher:** Stuck? Ask your teacher or a classmate for help. Two pairs of eyes are better than one!

Thank You & Keep Coding!

Great job! Keep exploring with Python Turtle. You are amazing coders — the journey has just begun! 🌟